Machine Learning in Healthcare

This paper presents an overview of machine learning applications in the healthcare domain, covering diagnosis, prognosis, treatment recommendation, and clinical workflow optimisation.

Introduction

Machine learning has emerged as a transformative technology in medicine. From early cancer detection to personalised drug dosing, the potential applications are vast and growing rapidly.

Motivation

The exponential growth of electronic health records, medical imaging, and wearable sensor data has created an opportunity for data-driven decision support systems.

Data Sources

Primary data sources include EHR systems, PACS imaging archives, insurance claims databases, and prospective clinical trial datasets.

Dataset Collection

We collected data from three hospital systems spanning five years. All data were de-identified in accordance with HIPAA regulations before being ingested into the training pipeline.

Preprocessing

Raw records were normalised, missing values were imputed using median substitution for continuous variables and mode imputation for categorical variables.

Feature Engineering

Temporal features were computed using a sliding window approach. Demographic variables were one-hot encoded. Lab values were standardised to z-scores.

Model Architecture

We evaluated three families of models: gradient-boosted trees, deep neural networks, and transformer-based architectures applied to sequential clinical data.

Transformer Encoder

The transformer encoder processes time-ordered event sequences with multi-head self-attention, allowing the model to capture long-range dependencies between clinical events.

Positional Encoding

Time-aware positional encodings encode the actual elapsed time between events rather than using fixed sinusoidal encodings.

Results

The transformer model achieved an AUROC of 0.923 on the held-out test set, outperforming the gradient-boosted baseline by 4.1 percentage points and the deep neural network by 2.6 points.

Conclusion

Transformer-based models show strong promise for clinical risk stratification. Future work will explore federated learning to train across institutions without sharing raw patient data.